

# CHAPTER 1: ALGEBRAIC METHODS

This chapter forms the foundation for many high-level mathematical techniques. You will develop fluency in manipulating algebraic expressions, dividing polynomials, and applying the factor and remainder theorems. In addition, you will explore the nature of mathematical proof through different methods including deduction, counter-examples, and proof by exhaustion.

## 1.1 Algebraic Fractions

### A. Simplifying Algebraic Fractions

**Definition:** An algebraic fraction is a fraction where the numerator and/or denominator are algebraic expressions (typically polynomials). The goal is to rewrite the expression in its simplest form.

#### Steps for Simplification:

1. **Factorise** the numerator and the denominator separately.
2. **Cancel common factors**, ensuring that any restrictions on the variable (to avoid division by zero) are noted.
3. **Simplify the expression** to its reduced form.

#### Example 1: Simplify

$$\frac{x^2 - 9}{x^2 - 4x + 3}$$

#### Solution:

1. Factorise numerator and denominator.
  - Numerator:  $x^2 - 9$  is a difference of two squares:

$$x^2 - 9 = (x - 3)(x + 3).$$

- Denominator:  $x^2 - 4x + 3$  factors as:

$$x^2 - 4x + 3 = (x - 1)(x - 3).$$

2. Write the fraction as:

$$\frac{(x - 3)(x + 3)}{(x - 1)(x - 3)}.$$

3. Cancel the common factor  $(x - 3)$  (provided  $x \neq 3$ ):

$$= \frac{x+3}{x-1}$$

4. **Result:**

$$\frac{x^2 - 9}{x^2 - 4x + 3} = \frac{x+3}{x-1}, \quad x \neq 3, x \neq 1.$$

**Example 2:** Simplify

$$\frac{2x^2 - 8}{4x^2 - 16x}.$$

**Solution:**

1. Factorise numerator and denominator.
  - Numerator:  $2x^2 - 8 = 2(x^2 - 4) = 2(x-2)(x+2)$ .
  - Denominator:  $4x^2 - 16x = 4x(x-4)$ .
2. Write the expression as:

$$\frac{2(x-2)(x+2)}{4x(x-4)}.$$

3. Simplify constants:  $\frac{2}{4} = \frac{1}{2}$ , so we have

$$= \frac{(x-2)(x+2)}{2x(x-4)}.$$

4. **Result:**

$$\frac{2x^2 - 8}{4x^2 - 16x} = \frac{(x-2)(x+2)}{2x(x-4)}, \quad x \neq 0, x \neq 4.$$

## 1.2 Dividing Polynomials

Dividing polynomials involves rewriting a polynomial (dividend) as the product of a divisor and a quotient, plus a remainder.

### A. Understanding a Polynomial

A **polynomial** is an expression of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where the coefficients  $a_n, a_{n-1}, \dots, a_0$  are real numbers (or from another field), and  $n$  is a non-negative integer.

## B. Dividing a Polynomial by a Linear Expression Where the Coefficient of $x$ is 1

When the divisor is in the form  $x - p$ , the process is often simpler. One approach is **synthetic division** (or simply polynomial division when arranged properly).

**Example 3:** Divide  $x^3 + 2x^2 - 5x + 6$  by  $x - 2$ .

**Solution:**

1. **Set up synthetic division** with  $p = 2$  (since  $x - 2 = 0 \rightarrow x = 2$ ).
2. Write the coefficients: 1, 2, -5, 6.
3. Perform synthetic division:
  - Bring down 1.
  - Multiply 1 by 2, get 2; add to 2  $\rightarrow$  4.
  - Multiply 4 by 2, get 8; add to -5  $\rightarrow$  3.
  - Multiply 3 by 2, get 6; add to 6  $\rightarrow$  12.
4. **Interpretation:** The quotient is  $x^2 + 4x + 3$  and the remainder is 12. In notation:

$$x^3 + 2x^2 - 5x + 6 = (x - 2)(x^2 + 4x + 3) + 12.$$

*Note:* If you expect no remainder (for factorization purposes), adjust or check your work; here the remainder theory is highlighted.

## C. Using Long Division for Dividing a Polynomial by $(x \pm p)$

Long division is similar to numerical long division.

**Example 4:** Divide  $2x^3 - 3x^2 + 4x - 5$  by  $x + 1$ .

**Steps:**

1. Write the division in long division form:

$$2x^3 - 3x^2 + 4x - 5 \text{ by } x + 1$$

2. **Divide**  $2x^3$  by  $x$  to get  $2x^2$ . Multiply the divisor by  $2x^2$ :

$$2x^2 \cdot (x + 1) = 2x^3 + 2x^2.$$

3. **Subtract:**

$$(2x^3 - 3x^2) - (2x^3 + 2x^2) = -5x^2.$$

4. **Bring down the next term:**  $+4x$  gives  $-5x^2 + 4x$ .
5. **Divide**  $-5x^2$  by  $x$  to get  $-5x$ . Multiply the divisor by  $-5x$ :

$$-5x \cdot (x + 1) = -5x^2 - 5x.$$

6. **Subtract:**

$$(-5x^2 + 4x) - (-5x^2 - 5x) = 9x.$$

7. **Bring down the constant -5 to get:**  $9x - 5$ .
8. **Divide**  $9x$  by  $x$  to get  $9$ . Multiply:

$$9(x + 1) = 9x + 9.$$

9. **Subtract:**

$$(9x - 5) - (9x + 9) = -14.$$

10. **Result:** Quotient:  $2x^2 - 5x + 9$ ; Remainder:  $-14$ .

Thus,

$$2x^3 - 3x^2 + 4x - 5 = (x + 1)(2x^2 - 5x + 9) - 14.$$

## D. Finding the Remainder When a Polynomial is Divided by a Linear Expression

The **Remainder Theorem** (discussed in Section 1.4) states that the remainder when  $f(x)$  is divided by  $x - p$  is simply  $f(p)$ .

**Example 5:** Find the remainder when  $f(x) = 3x^3 + x^2 - 8x + 5$  is divided by  $x - 2$ .

**Solution:** Apply the Remainder Theorem: Evaluate  $f(2)$ :

$$f(2) = 3(2)^3 + (2)^2 - 8(2) + 5 = 3 \cdot 8 + 4 - 16 + 5 = 24 + 4 - 16 + 5 = 17.$$

Thus, the remainder is 17.

## 1.3 The Factor Theorem

### A. Understanding and Using the Factor Theorem

**Statement:**

- **Forward:** If  $f(p) = 0$  for a polynomial  $f(x)$ , then  $(x - p)$  is a factor of  $f(x)$ .
- **Converse:** If  $(x - p)$  is a factor of  $f(x)$ , then  $f(p) = 0$ .

## B. Extended Factor Theorem for Rational Roots

- If  $f\left(\frac{q}{p}\right) = 0$ , then  $(px - q)$  is a factor of  $f(x)$ .

## C. Examples

**Example 6:** Let  $f(x) = x^3 - 4x^2 + x + 6$ . Verify whether  $(x - 2)$  is a factor.

**Solution:** Compute  $f(2)$ :

$$f(2) = 2^3 - 4 \cdot 2^2 + 2 + 6 = 8 - 16 + 2 + 6 = 0.$$

Thus, since  $f(2) = 0$ ,  $(x - 2)$  is a factor of  $f(x)$ .

**Example 7:** Factorise the cubic polynomial  $f(x) = 2x^3 + 3x^2 - 8x - 12$  using the factor theorem.

**Steps:**

1. Look for a root by testing simple candidates (e.g.,  $x = 1, -1, 2, -2, 3, -3$ , etc.).
2. Try  $x = 2$ :

$$f(2) = 2(8) + 3(4) - 8(2) - 12 = 16 + 12 - 16 - 12 = 0.$$

So,  $x = 2$  is a root and hence  $(x - 2)$  is a factor.

3. Divide  $f(x)$  by  $(x - 2)$  (synthetic or long division) to get a quadratic factor.
  - The division yields a quadratic  $2x^2 + 7x + 6$ .
4. Factorise the quadratic:

$$2x^2 + 7x + 6 = (2x + 3)(x + 2).$$

5. Therefore, the complete factorisation is:

$$f(x) = (x - 2)(2x + 3)(x + 2).$$

## D. Remarks on Factor Theorem

- The factor theorem is a powerful tool in factorising polynomials, especially higher degree ones, by reducing them into linear and quadratic factors.
- It also leads to the identification of all real roots of a polynomial.

## 1.4 The Remainder Theorem

## A. Understanding and Using the Remainder Theorem

**Statement:** If a polynomial  $f(x)$  is divided by a linear divisor of the form  $(ax \pm b)$ , then the remainder is given by  $f\left(\mp \frac{b}{a}\right)$ .

## B. Examples

**Example 8:** Find the remainder when  $f(x) = 4x^3 - 6x^2 + 3x - 5$  is divided by  $2x + 1$ .

**Solution:**

1. Write the divisor in the form  $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$ .
2. Evaluate  $f\left(-\frac{1}{2}\right)$ :

$$f\left(-\frac{1}{2}\right) = 4\left(-\frac{1}{2}\right)^3 - 6\left(-\frac{1}{2}\right)^2 + 3\left(-\frac{1}{2}\right) - 5.$$

Calculate each term:

- $\left(-\frac{1}{2}\right)^3 = -\frac{1}{8}$  so  $4 \cdot \left(-\frac{1}{8}\right) = -\frac{1}{2}$ .
- $\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$  so  $-6 \cdot \frac{1}{4} = -\frac{3}{2}$ .
- $3\left(-\frac{1}{2}\right) = -\frac{3}{2}$ .

Sum:

$$-\frac{1}{2} - \frac{3}{2} - \frac{3}{2} - 5 = -\frac{1+3+3}{2} - 5 = -\frac{7}{2} - 5 = -\frac{7+10}{2} = -\frac{17}{2}.$$

3. **Result:** The remainder is  $-\frac{17}{2}$ .

## 1.5 Mathematical Proof

Mathematical proof is the backbone of mathematics. It involves establishing the truth of a statement through logical reasoning and deduction.

### A. Understanding the Structure of a Mathematical Proof

- **Key Elements:**
  - **Statement/Claim:** What you need to prove.
  - **Given Information:** The hypotheses or axioms that are assumed true.

- **Logical Deduction:** A sequence of steps leading from the given information to the claim.
- **Conclusion:** A final statement that confirms the claim.

## B. Using Deduction to Show a Statement is True

- **Deductive Reasoning:** Begin with known facts and axioms, then apply logical steps to arrive at the conclusion.

**Example 9 (Simple Proof):** Prove that the sum of two even numbers is even.

**Proof:**

1. Let  $a$  and  $b$  be two even numbers. By definition, there exist integers  $m$  and  $n$  such that:

$$a = 2m \quad \text{and} \quad b = 2n.$$

2. Then,

$$a + b = 2m + 2n = 2(m + n).$$

3. Since  $m + n$  is an integer,  $a + b$  is divisible by 2.
4. **Conclusion:** The sum of two even numbers is even.

## C. Proving and Disproving Statements Using Examples

- **Method:**
  - To **prove a statement true**, provide a general proof (as above).
  - To **disprove a statement**, it is sufficient to find a counter-example.

**Example 10 (Counter-Example):** Disprove the statement “All prime numbers are odd.”

- **Counter-Example:** 2 is a prime number (divisible only by 1 and itself) but is even.
- **Conclusion:** The statement is false.

## D. Proof by Exhaustion

This method involves verifying the statement for every possible case, often used when the number of cases is finite.

**Example:** To prove that every positive integer less than 5 is either 1, 2, 3, or 4, list and check each possibility.

- **Verification:** Clearly, the only possibilities when  $1 \leq n < 5$  are  $n = 1, 2, 3, 4$ .
- **Conclusion:** Statement is established by exhausting all cases.

## 1.6 Methods of Proof

Different methods of proof offer various ways to establish mathematical truths.

### A. Proving Identities

- **Example:** Prove that for any real number  $x$ ,

$$(x + 1)^2 = x^2 + 2x + 1.$$

**Proof:** Expand the left-hand side:

$$(x + 1)^2 = (x + 1)(x + 1) = x \cdot x + x \cdot 1 + 1 \cdot x + 1 \cdot 1 = x^2 + 2x + 1.$$

Thus, the identity is verified.

### B. Disproving Statements Using Counter-examples

- **Method:** Identify a single example where the statement fails.

**Example:** To disprove “All functions are continuous,” simply consider the function

$$f(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0. \end{cases}$$

At  $x = 0$ , the function has a jump discontinuity. Hence, the statement is false.

### C. Proof by Exhaustion

- **Definition:** Break the proof into a finite number of cases and show the statement holds for each.
- **Example:** Prove that if  $n$  is an integer with  $0 \leq n \leq 4$ , then  $n^2 < 20$ . Check each case:
  - $n = 0$ :  $0^2 = 0 < 20$ .
  - $n = 1$ :  $1^2 = 1 < 20$ .
  - $n = 2$ :  $2^2 = 4 < 20$ .
  - $n = 3$ :  $3^2 = 9 < 20$ .
  - $n = 4$ :  $4^2 = 16 < 20$ .

Since all cases satisfy the condition, the statement is proven.

## CHAPTER REVIEW 1

Let's briefly summarize the major points:

- **Algebraic Fractions:**

- Simplification involves factorizing both numerator and denominator and cancelling common factors.
- **Dividing Polynomials:**
  - Understand the process of synthetic division and long division, and use the Remainder Theorem as a shortcut: The remainder when dividing by  $x-p$  is  $f(p)$ .
- **The Factor Theorem:**
  - A polynomial  $f(x)$  has  $(x-p)$  as a factor if and only if  $f(p)=0$ . This extends to rational roots where if  $f\left(\frac{q}{p}\right) = 0$  then  $(px - q)$  is a factor.
- **The Remainder Theorem:**
  - When dividing  $f(x)$  by  $(ax \pm b)$ , the remainder is  $f\left(\mp \frac{b}{a}\right)$ .
- **Mathematical Proof:**
  - Proofs require a clear statement, logical deduction, and use of examples or counter-examples. Proof by exhaustion confirms a statement by checking each possible case.
- **Methods of Proof:**
  - Proving identities, disproving claims with counter-examples, and using proof by exhaustion are all essential techniques in rigorous mathematics.

## SUMMARY OF KEY POINTS

- **Simplifying Algebraic Fractions** requires factorisation and cancellation while respecting conditions on the variable.
- **Dividing Polynomials** can be achieved by synthetic division or long division, with the Remainder Theorem providing an efficient shortcut.
- **The Factor Theorem** ensures that if a polynomial equals zero at a certain point, the corresponding linear factor divides the polynomial.
- **The Remainder Theorem** tells us how to quickly find the remainder when dividing by a linear expression.
- **Mathematical Proof** is a structured process employing deduction, counter-examples, and exhaustive checking to establish or refute claims.

## CHAPTER 2: COORDINATE GEOMETRY IN THE (x, y) PLANE

This chapter focuses on some of the central topics in coordinate geometry. You will learn how to work with line segments, circles, and their interactions. In particular, mastering midpoints, perpendicular bisectors, and the different forms of a circle's equation will lay the groundwork for exploring intersections and the geometric properties of tangents, chords, and triangles in the plane.

## 2.1 Midpoints and Perpendicular Bisectors

### A. Finding the Midpoint of a Line Segment

**Definition:** The midpoint of a line segment joining two points  $(x_1, y_1)$  and  $(x_2, y_2)$  is the point exactly halfway between them. It is given by the formula:

$$\text{Midpoint } M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

**Example 1:** Find the midpoint of the segment joining  $(3, 4)$  and  $(7, 10)$ .

**Solution:**

$$x\text{-coordinate: } \frac{3 + 7}{2} = \frac{10}{2} = 5, \quad y\text{-coordinate: } \frac{4 + 10}{2} = \frac{14}{2} = 7.$$

Thus,  $M = (5, 7)$ .

### B. Finding the Equation of the Perpendicular Bisector

**Definition:** A perpendicular bisector of a line segment is a line that:

- Passes through the midpoint of the segment.
- Is perpendicular to the line segment.

**Steps to Find the Equation:**

1. **Determine the Midpoint M:** Use the midpoint formula.
2. **Find the Gradient of the Line Segment:** For points  $(x_1, y_1)$  and  $(x_2, y_2)$ , the gradient is:

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

3. **Find the Gradient of the Perpendicular Bisector:** The perpendicular bisector's gradient,  $m_{\perp}$ , is the negative reciprocal of  $m$  (provided  $m \neq 0$ ):

$$m_{\perp} = -\frac{1}{m}.$$

If the line segment is vertical (undefined slope), then the bisector is horizontal (slope 0), and vice versa.

4. **Write the Equation Using the Point-Slope Form:** With midpoint  $M(x_m, y_m)$  and gradient  $m_{\perp}$ , the equation is:

$$y - y_m = m_{\perp}(x - x_m).$$

**Example 2:** For the same segment joining (3, 4) and (7, 10), find the equation of the perpendicular bisector.

**Solution:**

1. **Midpoint M:** We found  $M = (5, 7)$ .
2. **Gradient of the Segment:**

$$m = \frac{10 - 4}{7 - 3} = \frac{6}{4} = \frac{3}{2}.$$

3. **Gradient of Perpendicular Bisector:**

$$m_{\perp} = -\frac{1}{\frac{3}{2}} = -\frac{2}{3}.$$

4. **Equation (Point-Slope Form):**

$$y - 7 = -\frac{2}{3}(x - 5).$$

This is the required equation of the perpendicular bisector.

## 2.2 Equation of a Circle

Circles are fundamental loci in the Cartesian plane. Their equations describe all the points that are a fixed distance (the radius) from a fixed point (the centre).

### A. Circle with Centre (0, 0) and Radius r

**Standard Equation:** For a circle centered at the origin with radius r, all points (x, y) satisfy:

$$x^2 + y^2 = r^2.$$

**Example 3:** For  $r=5$ , the equation is:

$$x^2 + y^2 = 25.$$

### B. Circle with Centre (a, b) and Radius r

**Standard Equation:** For a circle centered at (a, b) with radius r, the relationship is:

$$(x - a)^2 + (y - b)^2 = r^2.$$

**Example 4:** For a circle with centre(3, -2) and  $r = 4$ , the equation is:

$$(x - 3)^2 + (y + 2)^2 = 16.$$

### C. Finding the Centre and Radius from the General Equation

A circle is sometimes given in the general form:

$$x^2 + y^2 + 2gx + 2fy + c = 0.$$

To find the centre and radius, complete the square for both xx and yy.

**Steps:**

**1. Group and Rearrange:**

$$(x^2 + 2gx) + (y^2 + 2fy) = -c.$$

**2. Complete the Square:**

- For x:  $x^2 + 2gx = (x + g)^2 - g^2$ .
- For y:  $y^2 + 2fy = (y + f)^2 - f^2$ .

**3. Rewrite Equation:**

$$(x + g)^2 + (y + f)^2 = g^2 + f^2 - c.$$

**4. Centre and Radius:**

- The centre is  $(-g, -f)$ .
- The radius is  $r = \sqrt{g^2 + f^2 - c}$  (provided  $g^2 + f^2 - c > 0$ ).

**Example 5:** Find the centre and radius for:

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

**Solution:**

**1. Group x and y terms:**

$$(x^2 - 6x) + (y^2 + 4y) = 12.$$

**2. Complete the square:**

- For x:  $x^2 - 6x = (x - 3)^2 - 9$ .
- For y:  $y^2 + 4y = (y + 2)^2 - 4$ .

**3. Rewrite:**

$$(x - 3)^2 - 9 + (y + 2)^2 - 4 = 12.$$

Combine constants:

$$(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25.$$

**4. Centre and Radius:**

- Centre: (3, -2).
- Radius:  $r = 5$ .

## 2.3 Intersections of Straight Lines and Circles

Finding the points where a line and a circle intersect involves solving a system of equations.

### A. Method Overview

**1. Write the Equations:**

- For the circle: for example,  $(x - a)^2 + (y - b)^2 = r^2$ .
- For the line: typically in the form  $y = mx + c$  (or another form).

**2. Substitute the Line Equation into the Circle Equation:** Replace  $y$  in the circle's equation with the expression from the line.

**3. Solve the Resulting Equation (Usually Quadratic) for  $x$  and Then  $y$ .**

### B. Example

**Example 6:** Find the intersection of the circle:

$$(x - 2)^2 + (y + 1)^2 = 16,$$

with the line:

$$y = 2x - 3.$$

#### **Solution:**

1. Substitute  $y = 2x - 3$  into the circle's equation:

$$(x - 2)^2 + ((2x - 3) + 1)^2 = 16.$$

Simplify:

$$(x - 2)^2 + (2x - 2)^2 = 16.$$

2. Expand both squares:

$$\begin{aligned}(x - 2)^2 &= x^2 - 4x + 4, \\ (2x - 2)^2 &= 4x^2 - 8x + 4.\end{aligned}$$

3. Combine:

$$(x^2 - 4x + 4) + (4x^2 - 8x + 4) = 5x^2 - 12x + 8 = 16.$$

4. Rearrange:

$$\begin{aligned} 5x^2 - 12x + 8 - 16 &= 0, \\ 5x^2 - 12x - 8 &= 0. \end{aligned}$$

5. Solve the quadratic using the quadratic formula:

$$x = \frac{12 \pm \sqrt{(-12)^2 - 4(5)(-8)}}{2 \cdot 5} = \frac{12 \pm \sqrt{144 + 160}}{10} = \frac{12 \pm \sqrt{304}}{10}.$$

Simplify  $\sqrt{304} = 4\sqrt{19}$  so:

$$x = \frac{12 \pm 4\sqrt{19}}{10} = \frac{6 \pm 2\sqrt{19}}{5}.$$

6. **Find y:** Substitute  $x$  back into  $y = 2x - 3$  for each value.

These values give the intersection points.

## 2.4 Use of Tangent and Chord Properties

### A. Tangent-Radius Perpendicularity

**Key Property:** A tangent to a circle is perpendicular to the radius drawn to the point of tangency.

**Example 7:** For a circle with centre  $(h, k)$  and a point of tangency  $(x_1, y_1)$ , if the tangent line's gradient is  $m_t$  and the radius' gradient is  $m_r$ , then:

$$m_t \cdot m_r = -1.$$

### B. Other Circle Theorems

While not all are detailed here, some important ones include:

- **Chord properties:** The perpendicular from the centre of a circle to a chord bisects the chord.
- **Angle properties:** The angle in a semicircle is a right angle; angles at the centre are twice the angle at the circumference subtended by the same arc.

**Usage:** These properties help solve problems where you are given parts of a circle (chords, tangents) and must find missing lengths or angles.

## 2.5 Circles and Triangles

### A. Solving Geometric Problems Involving Circles and Triangles

Problems often combine circle properties with triangle geometry. Examples of such applications include:

- **Finding the circumcentre of a triangle:** The circumcentre is the intersection of the perpendicular bisectors of the triangle's sides, and it is the centre of the circumscribed circle.
- **Inscribed and central angles:** Using the relationships between inscribed angles and intercepted arcs to solve for unknown angles.

**Example 8:** Suppose a triangle is inscribed in a circle, and you are given the lengths of the sides of the triangle. You might be asked to compute the radius of the circumscribed circle using the formula:

$$R = \frac{abc}{4\Delta},$$

where  $a$ ,  $b$ ,  $c$  are the side lengths and  $\Delta$  is the area of the triangle.

While a specific numerical example is not provided here, note that combining the Heron's formula (for area) with the above relationship allows one to find the circumradius of a triangle.

## CHAPTER REVIEW 2

### Key Topics Covered:

1. **Midpoints and Perpendicular Bisectors:**
  - Compute the midpoint using  $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ .
  - Find the perpendicular bisector by using the negative reciprocal of the segment's slope and passing through the midpoint.
2. **Equations of Circle:**
  - For a circle with centre  $(0,0)$ :  $x^2 + y^2 = r^2$ .
  - For a circle with centre  $(a,b)$ :  $(x - a)^2 + (y - b)^2 = r^2$ .
  - Complete the square to rewrite an equation in the form  $x^2 + y^2 + 2gx + 2fy + c = 0$  so that the centre and radius can be determined.
3. **Intersections of Lines and Circles:**
  - Solve by substituting the expression for  $y$  (or  $x$ ) from the line into the circle's equation.
  - Solve the resulting quadratic to determine the points of intersection.

#### 4. Tangent and Chord Properties:

- A tangent is perpendicular to the radius at the point of tangency.
- Other circle theorems (such as chord bisectors and angle relationships) are essential to solving geometry problems.

#### 5. Circles and Triangles:

- Many problems combine circle geometry (like circumcircles) with triangle properties to determine unknown lengths, angles, or areas.

## SUMMARY OF KEY POINTS

- **Midpoints:** The midpoint of  $(x_1, y_1)$  and  $(x_2, y_2)$  is  $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ .
- **Perpendicular Bisectors:** Have a slope that is the negative reciprocal of the line segment's slope and pass through the segment's midpoint.
- **Equations of Circles:**
  - $x^2 + y^2 = r^2$  if centred at the origin.
  - $(x - a)^2 + (y - b)^2 = r^2$  for any centre  $(a, b)$ .
  - Completing the square allows recovery of the centre and radius from a general circle equation.
- **Intersection of Lines and Circles:** Substituting the linear equation into the circle's equation yields a quadratic that can be solved for the intersection points.
- **Tangent and Chord Properties:** A tangent is perpendicular to the radius at the contact point, and many useful geometric relationships arise from chord and angle theorems.
- **Circles and Triangles:** Problems may involve finding the circumcentre of a triangle, using inscribed angle properties, or computing circumradii.

## CHAPTER 3: EXPONENTIALS AND LOGARITHMS

Exponentials and logarithms are inverse operations that provide powerful ways to model growth and decay, solve equations, and simplify complex expressions. In this chapter, you will learn to graph and analyze exponential functions, understand what logarithms are, master their laws, use them to solve equations, and change between bases conveniently.

### 3.1 Exponential Functions

#### A. Definition and General Form

- **Definition:** An exponential function is one in which the independent variable appears in the exponent. Its general form is:

$$y = a^x, \quad \text{where } a > 0 \text{ and } a \neq 1.$$

- **Key Concepts:**
  - When  $a > 1$ , the function represents exponential growth.
  - When  $0 < a < 1$ , the function represents exponential decay.
  - The function is always positive (i.e.  $y > 0$  for all  $x$ ).
  - The horizontal line  $y = 0$  is a horizontal asymptote.

## B. Sketching Graphs of Exponential Functions

### General Steps:

1. **Determine the key features:**
  - **Y-intercept:** Since  $a^0 = 1$ , the graph always passes through  $(0, 1)$ .
  - **Growth or Decay:**
    - If  $a > 1$ : the function increases as  $x$  increases.
    - If  $0 < a < 1$ : the function decreases as  $x$  increases.
2. **Plot additional points:** Choose values of  $x$  (for example,  $x = 1$  and  $x = -1$ ) and compute  $a^x$  to help shape the curve.
3. **Draw asymptote:** Clearly indicate  $y = 0$  as the asymptote.

**Example 1:** Sketch the graph of

$$y = 2^x.$$

- **Y-intercept:** When  $x = 0, y = 2^0 = 1$  (point  $(0,1)$ ).
- **Other points:**  $x = 1: y = 2^1 = 2$ ;  $x = -1: y = 2^{-1} = \frac{1}{2}$ .
- **Behaviour:** Since  $2 > 1$ , the function is increasing and has exponential growth.
- **Asymptote:**  $y = 0$  is approached as  $x \rightarrow -\infty$ .

*Sketch:* A smooth, continuously rising curve that passes through  $(0,1)$ ,  $(1,2)$ , and  $(-1,0.5)$  with the horizontal line  $y = 0$  as the lower bound.

## C. Understanding Properties of Exponential Functions

- **Domain and Range:** Domain: All real numbers,  $x \in (-\infty, \infty)$ . Range:  $y \in (0, \infty)$ .
- **Intercepts:** Always crosses the y-axis at  $y=1$  (if no vertical shift).
- **Growth/Decay Rate:** Determined by the base  $a$ ; larger  $a$  gives a steeper graph (for growth) and smaller  $0 < a < 1$  yields rapid decay.
- **Transformations:** Adding a constant outside the exponential (e.g.  $y = a^x + k$ ) results in a vertical shift; multiplying by a coefficient (e.g.  $y = c \cdot a^x$ ) scales the graph vertically.

## 3.2 Logarithms

### A. Definition of a Logarithm

- **Definition:** A logarithm is the inverse operation to exponentiation. For a positive real number  $a$  (with  $a \neq 1$ ) and a positive number  $x$ , the logarithm is defined by:

$$\log_a x = n \quad \text{if and only if} \quad a^n = x.$$

- **Interpretation:** The logarithm  $\log_a x$  is the exponent  $n$  to which the base  $a$  must be raised to yield  $x$ .

## B. Converting Between Exponential and Logarithmic Forms

- **Exponential Form:**  $a^n = x$ .
- **Logarithmic Form:**  $\log_a x = n$ .

**Example 2:** Convert  $3^4 = 81$  to logarithmic form.

- This becomes:

$$\log_3 81 = 4.$$

**Example 3:** Convert  $\log_5 25 = 2$  back to exponential form.

- This can be written as:

$$5^2 = 25.$$

## 3.3 Laws of Logarithms

Logarithms have several properties (or laws) that make them very useful for simplifying expressions and solving equations.

### A. Multiplication Law

- **Law:**

$$\log_a(xy) = \log_a x + \log_a y.$$

**Example 4:** Using base 2,

$$\log_2(8 \cdot 4) = \log_2 8 + \log_2 4.$$

Since  $\log_2 8 = 3$  and  $\log_2 4 = 2$ ,

$$\log_2 32 = 3 + 2 = 5.$$

### B. Division Law

- **Law:**

$$\log_a \frac{x}{y} = \log_a x - \log_a y.$$

**Example 5:** Using base 10,

$$\log_{10} \left( \frac{100}{10} \right) = \log_{10} 100 - \log_{10} 10 = 2 - 1 = 1.$$

### C. Power Law

- **Law:**

$$\log_a x^k = k \log_a x.$$

**Example 6:** Calculate  $\log_3 27^2$ . Since  $27 = 3^3$ ,

$$\log_3 27^2 = 2 \log_3 27 = 2 \cdot 3 = 6.$$

### D. Additional Logarithmic Properties

- **Reciprocal Property:**

$$\log_a \frac{1}{x} = -\log_a x.$$

- **Identity for 1:**

$$\log_a 1 = 0 \quad \text{since } a^0 = 1.$$

- **Self-base Identity:**

$$\log_a a = 1 \quad \text{since } a^1 = a.$$

**Example 7:** Show that  $\log_5 \frac{1}{125} = -3$ : Since  $125 = 5^3$

## 3.4 Solving Equations Using Logarithms

Logarithms are instrumental in solving both exponential and logarithmic equations.

### A. Solving Exponential Equations Using Logarithms

**Example 8:** Solve  $2^x = 10$ .

**Steps:**

1. Take the logarithm of both sides  $\ln 2^x = \ln 10$ .
2. Use the power law:

$$x \ln 2 = \ln 10.$$

3. Solve for x:

$$x = \frac{\ln 10}{\ln 2}.$$

**B. Solving Logarithmic Equations**

**Example 9:** Solve  $\log_3(x - 1) = 2$ .

**Steps:**

1. Convert to exponential form:

$$3^2 = x - 1.$$

2. Compute and solve:

$$9 = x - 1 \quad \Rightarrow \quad x = 10.$$

*Note:* When solving logarithmic equations, always check that the arguments of all logarithms are positive.

## 3.5 Changing the Base of a Logarithm

**A. The Change of Base Formula**

If you wish to compute a logarithm with one base using logarithms of another base, use:

$$\log_b x = \frac{\log_a x}{\log_a b},$$

where a is any positive number (commonly  $a = 10$  or  $a = e$ ).

**Example 10:** Express  $\log_2 32$  in terms of common logarithms (base 10):

Since  $32 = 2^5$ , we have

$$\log_2 32 = \frac{\log_{10} 32}{\log_{10} 2}.$$

Numerically, if needed, you could compute those values using a calculator.

## CHAPTER REVIEW 3

Let's consolidate the key topics of this chapter:

1. **Exponential Functions:**
  - The function  $y = a^x$  (with  $a > 0$  and  $a \neq 1$ ) represents exponential growth (if  $a > 1$ ) or decay (if  $0 < a < 1$ ).
  - Key features include the y-intercept at (0,1) and the horizontal asymptote  $y = 0$ .
2. **Logarithms:**
  - Defined as the inverse of exponentiation, with  $\log_a x = n$  if and only if  $a^n = x$ .
  - Conversions between exponential and logarithmic forms are essential.
3. **Laws of Logarithms:**
  - Multiplication, division, and power laws simplify the process of combining and breaking apart logarithmic expressions.
  - Additional properties such as  $\log_a 1 = 0$  and  $\log_a a = 1$  are fundamental to simplifying expressions.
4. **Solving Equations:**
  - Exponential equations can be solved using logarithms to “bring down” the exponent.
  - Logarithmic equations are turned into exponential form to isolate the variable.
5. **Changing the Base:**
  - The change of base formula enables the evaluation of logarithms in a more convenient base.

## SUMMARY OF KEY POINTS

- **Exponential Functions:**  $y = a^x$  models growth and decay with  $y > 0$  for all  $x$ , a y-intercept at 1, and a horizontal asymptote at 0.
- **Logarithms:** They are defined by the equivalence  $\log_a x = n \Leftrightarrow a^n = x$ . Converting between forms is essential.
- **Logarithm Laws:**
  - Multiplication Law:  $\log_a(xy) = \log_a x + \log_a y$ .
  - Division Law:  $\log_a \frac{x}{y} = \log_a x - \log_a y$ .
  - Power Law:  $\log_a x^k = k \log_a x$ . Also,  $\log_a \frac{1}{x} = -\log_a x$ ,  $\log_a 1 = 0$ , and  $\log_a a = 1$ .
- **Solving Equations:** Both exponential and logarithmic equations can be solved by appropriate use of logarithms and conversion between the two forms.

- **Changing Base:** The formula  $\log_b x = \frac{\log_a x}{\log_a b}$  allows for evaluation in any convenient base.

## CHAPTER 4: THE BINOMIAL EXPANSION

The binomial expansion is a method of expressing a power of a binomial as a sum of terms involving binomial coefficients. This chapter covers not only how to expand expressions of the form  $(a + b)^n$  but also how to find specific terms, estimate expressions when  $xx$  is small, and understand the underlying structures (such as Pascal's triangle and factorial notation) that make the binomial theorem work.

### 4.1 Pascal's Triangle

Pascal's triangle is a triangular arrangement of numbers where each number is the sum of the two immediately above it. It directly provides the coefficients (often called binomial coefficients) used in the expansion of  $(a + b)^n$ .

#### A. Using Pascal's Triangle to Find Binomial Coefficients

- **Construction:**
  - **Row 0:** Starts with a single 1.
  - **Row 1:** Consists of 1, 1.
  - **Row 2:** 1, 2, 1.
  - **Row 3:** 1, 3, 3, 1.
  - And so on. In general, Row  $n$  contains  $n+1$  entries.
- **Example:** To expand  $(a + b)^3$ , reference Row 3:

1,3,3,1.1,\quad 3,\quad 3,\quad 1.

Thus,

$$(a + b)^3 = 1 \cdot a^3 + 3 \cdot a^2b + 3 \cdot ab^2 + 1 \cdot b^3.$$

#### B. Understanding the Relationship Between Pascal's Triangle and Binomial Expansion

- **Key Point:** The  $r^{\text{th}}$  entry (starting at  $r = 0$ ) in the  $n^{\text{th}}$  row corresponds to the binomial coefficient

$$\binom{n}{r}.$$

- **Interpretation:** When you expand  $(a + b)^n$ , the coefficient of the term  $a^{n-r}b^r$  is exactly  $\binom{n}{r}$ . For example, in  $(a + b)^4$ , Row 4 of Pascal's triangle gives 1, 4, 6, 4, 1, matching the coefficients:

$$[(a + b)^4 = \binom{4}{0}a^4 + \binom{4}{1}a^3b + \binom{4}{2}a^2b^2 + \binom{4}{3}ab^3 + \binom{4}{4}b^4.]$$

## 4.2 Factorial Notation

Factorials and binomial coefficient formulas formalize the numbers seen in Pascal's triangle.

### A. Understanding and Using Factorial Notation

- **Definition:** For any positive integer  $n$ ,

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1.$$

By convention,  $0! = 1$ .

- **Example:**  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ .

### B. Formula for Binomial Coefficients

- **Formula:**

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad \text{for } 0 \leq r \leq n.$$

- **Alternative Notation:** The binomial coefficient is also denoted as  ${}^nC_r$ .
- **Example:** Calculate  $\binom{5}{2}$ :

$$\binom{5}{2} = \frac{5!}{2!(5-2)!} = \frac{120}{2 \times 6} = \frac{120}{12} = 10.$$

## 4.3 The Binomial Expansion

The binomial theorem provides a formula to expand  $(a + b)^n$  for any positive integer  $n$ .

### A. The Binomial Theorem

- **Statement:**

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r,$$

or equivalently,

$$[(a + b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \cdots + \binom{n}{n} b^n.]$$

- **Key Insight:** There are exactly  $n+1$  terms in the expansion.

## B. Example of a Binomial Expansion

**Example:** Expand  $(a + b)^4$ .

1. **Write the coefficients** (from Pascal's Triangle, Row 4): 1, 4, 6, 4, 1.
2. **Apply theorem:**

$$[(a + b)^4 = 1 \cdot a^4 + 4 \cdot a^3 b + 6 \cdot a^2 b^2 + 4 \cdot a b^3 + 1 \cdot b^4.]$$

## 4.4 Solving Binomial Problems

Solving problems involving binomial expansions often requires finding specific terms or the coefficient of a particular power.

### A. Finding Specific Terms in a Binomial Expansion

- **General Term Formula (Term  $r+1$ ):**

$$[T_{r+1} = \binom{n}{r} a^{n-r} b^r.]$$

**Example 1:** Find the 4th term in the expansion of  $(2x - 3)^5$ .

1. Recognize  $n = 5$  and we want the 4th term, so use  $r = 3$  (since term numbering starts at  $r = 0$ ).
2. Substitute into the general term:

$$T_4 = \binom{5}{3} (2x)^{5-3} (-3)^3.$$

3. Compute each part:
  - $\binom{5}{3} = 10$ .
  - $(2x)^2 = 4x^2$ .
  - $(-3)^3 = -27$ .
4. Multiply:

$$T_4 = 10 \times 4x^2 \times (-27) = -1080x^2.$$

5. **Result:** The 4th term is  $-1080x^2$ .

## B. Finding the Coefficient of a Particular Power of $x$

Sometimes problems ask: “Find the coefficient of  $x^k$  in the expansion of  $(px + q)^n$ .”

**Example 2:** Find the coefficient of  $x^3$  in  $(3x + 2)^6$ .

1. **General term:**

$$[T_{r+1} = \binom{6}{r} (3x)^{6-r} (2)^r.]$$

2. We seek  $x^3$ . In  $(3x)^{6-r}$ , the exponent of  $x$  is  $6 - r$ . Set:

$$6 - r = 3 \quad \Rightarrow \quad r = 3.$$

3. Substitute  $r = 3$ :

$$[T_4 = \binom{6}{3} (3x)^3 (2)^3.]$$

4. Compute:

- $\binom{6}{3} = 20$ .
- $(3x)^3 = 27x^3$ .
- $2^3 = 8$ .

5. Multiply the numerical factors:

$$20 \times 27 \times 8 = 20 \times 216 = 4320.$$

6. **Result:** The coefficient of  $x^3$  is 4320.

## 4.5 Binomial Estimation

Binomial estimation uses the binomial expansion to obtain approximate values when  $|x|$  is small.

### A. The Concept of Binomial Approximation

If  $|x|$  is small, for any real number  $n$ , the expansion of  $(1+x)^n$  can be approximated by the first few terms:

$$(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2 + \dots$$

The approximation becomes better as  $x$  tends to 0.

## B. Example of Binomial Estimation

**Example:** Approximate  $(1.02)^3$ .

1. Write as:

$$(1+x)^n \quad \text{with } x = 0.02 \text{ and } n = 3.$$

2. Use the first few terms:

$$(1.02)^3 \approx 1 + 3(0.02) + \frac{3(3-1)}{2}(0.02)^2.$$

3. Calculate each term:

- 1
- $3(0.02) = 0.06.$
- $\frac{3 \times 2}{2}(0.0004) = 3 \times 0.0004 = 0.0012.$

4. Add up:

$$1 + 0.06 + 0.0012 \approx 1.0612.$$

5. **Result:**  $(1.02)^3$  is approximately 1.0612.

## CHAPTER REVIEW 4

### Key Points Recap:

1. **Pascal's Triangle:**

- Provides binomial coefficients.
- Each row corresponds to the expansion of  $(a+b)^n$ .

2. **Factorial Notation and Binomial Coefficients:**

- $n! = n(n-1)(n-2) \cdots 2 \cdot 1$  with  $0! = 1$ .
- Binomial coefficients are computed as:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

3. **The Binomial Expansion:**

- $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r.$
- There are  $n+1$  terms in the expansion.

#### 4. Solving Binomial Problems:

- Use the general term  $T_{r+1} = \binom{n}{r} a^{n-r} b^r$  to find specific terms and coefficients.

#### 5. Binomial Estimation:

- When  $xx$  is small in  $(1 + x)^n$ , the expansion can be approximated by the first few terms for a quick estimation.

## SUMMARY OF KEY POINTS

- **Pascal's Triangle** is a quick reference for binomial coefficients.
- **Factorial Notation** is essential for expressing binomial coefficients:

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

- **Binomial Expansion** provides a formula to expand  $(a + b)^n$  into  $n+1$  terms.
- **Specific term-finding** and **coefficient extraction** are handled via the general term formula.
- **Binomial Estimation** helps approximate expressions for small values of  $xx$ .

## CHAPTER 5: SEQUENCES AND SERIES

Sequences are ordered lists of numbers defined by a specific rule. Series are the sums of the terms of a sequence. Two of the most common families of sequences and series are arithmetic (linear) and geometric (exponential) ones. In this chapter, you will learn to identify, work with, and sum both types. You will also see how to use sigma ( $\Sigma$ ) notation, solve recurrence relations, and apply these ideas to real-world modelling.

### 5.1 Arithmetic Sequences

#### A. Identifying and Working with Arithmetic Sequences

**Definition:** An arithmetic sequence (or arithmetic progression) is a sequence of numbers in which the difference between consecutive terms is constant. This constant difference is called the *common difference*, denoted by  $dd$ .

- **General Form:** If the first term of an arithmetic sequence is  $aa$  and the common difference is  $dd$ , then the sequence is:

$$a, \quad a + d, \quad a + 2d, \quad a + 3d, \quad \dots$$

#### Properties:

- The difference between any term and the previous term is always  $dd$ .
- If  $d > 0$ , the sequence is increasing; if  $d < 0$ , it is decreasing.

**Example 1:** If  $a = 3$  and  $d = 2$ , then the sequence is:

$$3,; 3 + 2 = 5,; 5 + 2 = 7,; 7 + 2 = 9,; \dots$$

Thus, the sequence is  $3, 5, 7, 9, \dots$ .

## B. The $n$ th Term of an Arithmetic Sequence

The formula for the  $n$ th term of an arithmetic sequence is key to finding any term in the progression without listing all previous terms:

$$u_n = a + (n - 1)d,$$

where

- $u_n$  (or sometimes denoted  $a_n$ ) is the  $n$ th term,
- $a$  is the first term,
- $d$  is the common difference, and
- $n$  is the position of the term in the sequence.

**Example 2:** For the arithmetic sequence with  $a = 3$  and  $d = 2$ :

- The 10th term is

$$u_{10} = 3 + (10 - 1) \times 2 = 3 + 18 = 21.$$

## 5.2 Arithmetic Series

### A. Finding the Sum of the First $n$ Terms of an Arithmetic Series

An arithmetic series is the sum of the terms of an arithmetic sequence.

- **Definition:** If you add the first  $n$  terms of an arithmetic sequence, you get an arithmetic series with sum denoted by  $S_n$ .

### B. Formulas for the Sum of an Arithmetic Series

There are two equivalent formulas for the sum of the first  $n$  terms:

1. **Using First Term and Common Difference:**

$$S_n = \frac{n}{2} [2a + (n - 1)d].$$

2. **Using First Term and Last Term:**

$$S_n = \frac{n}{2}(a + l),$$

where  $l$  is the last term (which can be computed as  $l = a + (n - 1)d$ ).

**Example 3:** Find the sum of the first 10 terms of the arithmetic sequence with  $a = 3$  and  $d = 2$ .

- First, find the 10th term:

$$l = 3 + (10 - 1) \times 2 = 3 + 18 = 21.$$

- Then, use the second formula:

$$S_{10} = \frac{10}{2}(3 + 21) = 5 \times 24 = 120.$$

Thus, the sum is 120.

## 5.3 Geometric Sequences

### A. Identifying and Working with Geometric Sequences

**Definition:** A geometric sequence (or geometric progression) is a sequence where each term is obtained by multiplying the previous term by a constant called the *common ratio*, denoted by  $r$ .

- **General Form:** If the first term is  $a$  and the common ratio is  $r$ , then the sequence is:

$$a, \quad ar, \quad ar^2, \quad ar^3,$$

#### Properties:

- The ratio between consecutive terms is always  $r$ :

$$\frac{u_{n+1}}{u_n} = r.$$

- Depending on  $r$ :
  - If  $|r| > 1$ : the sequence grows.
  - If  $|r| < 1$ : the sequence decays.
  - If  $r$  is negative, the terms alternate in sign.

**Example 4:** For a geometric sequence with  $a = 2$  and  $r = 3$ : The sequence is 2, 6, 18, 54, ....

### B. The $n$ th Term of a Geometric Sequence

The formula for the  $n$ th term is:

$$u_n = a \cdot r^{n-1}.$$

**Example 5:** For the above sequence, the 5th term is:

$$u_5 = 2 \cdot 3^{5-1} = 2 \cdot 3^4 = 2 \cdot 81 = 162.$$

## 5.4 Geometric Series

### A. Finding the Sum of the First n Terms of a Geometric Series

A geometric series is the sum of the terms of a geometric sequence.

- **Definition:** The sum of the first n terms of a geometric series is denoted by  $S_n$ .

### B. Formula for the Sum of a Geometric Series

For  $r \neq 1$ , the sum is given by:

$$S_n = \frac{a(1 - r^n)}{1 - r}.$$

When  $r > 1$ , sometimes the formula is written as:

$$S_n = \frac{a(r^n - 1)}{r - 1},$$

since the denominator is then positive.

**Example 6:** Find the sum of the first 4 terms of the geometric series 2, 6, 18, 54 (here  $a=2$  and  $r=3$ ).

Using the formula:

$$S_4 = \frac{2(1 - 3^4)}{1 - 3} = \frac{2(1 - 81)}{-2} = \frac{2(-80)}{-2} = 80.$$

## 5.5 Sum to Infinity

### A. Understanding the Concept of the Sum to Infinity

For a geometric series to have a finite sum as  $n$  tends to infinity, the absolute value of the common ratio must be less than 1 ( $|r| < 1$ ). In that case, the series converges.

### B. Condition for Convergence

The sum to infinity exists only if

$$|r| < 1.$$

### C. Formula for the Sum to Infinity

If the condition is met, the sum to infinity is:

$$S_{\infty} = \frac{a}{1 - r}.$$

**Example 7:** Find the sum to infinity for a geometric series where  $a = 5$  and  $r = 0.5$ .

Calculation:

$$S_{\infty} = \frac{5}{1 - 0.5} = \frac{5}{0.5} = 10.$$

## 5.6 Sigma Notation

### A. Understanding and Using Sigma Notation

**Definition:** Sigma notation (the summation symbol  $\Sigma$ ) provides a compact way to denote the sum of a sequence of terms.

- **Form:**

$$\sum_{i=m}^n a_i,$$

which means add the terms  $a_m, a_{m+1}, \dots, a_n$ .

### B. Examples

**Example 8:** Express the sum of the first 5 positive integers using sigma notation.

$$1 + 2 + 3 + 4 + 5 = \sum_{i=1}^5 i.$$

**Example 9:** Write the sum of the terms in an arithmetic sequence with first term  $a$  and common difference  $d$  for  $n$  terms:

$$S_n = \sum_{i=1}^n [a + (i - 1)d].$$

Sigma notation is particularly useful when working with series formulas and proofs.

## 5.7 Recurrence Relations

### A. Generating Terms of a Sequence Using a Recurrence Relation

**Definition:** A recurrence relation defines each term of a sequence as a function of its preceding term(s).

- **Example (Arithmetic Recurrence):** If  $u_1 = 3$  and  $u_{n+1} = u_n + 2$ , the sequence is:

$$3, 5, 7, 9, \dots$$

### B. Finding a Specific Term in a Sequence Defined by a Recurrence Relation

Often, you can either use iterative methods or find an explicit formula.

**Example 10:** For the recurrence relation  $u_1 = 3$  and  $u_{n+1} = u_n + 2$ , find the 8th term.

- Recognize this is an arithmetic sequence with  $a = 3$  and  $d = 2$ . Use the explicit formula:

$$u_8 = 3 + (8 - 1) \times 2 = 3 + 14 = 17.$$

Alternatively, compute  $u_2 = 5, u_3 = 7, \dots$  until  $u_8 = 17$ .

## CHAPTER REVIEW 5

Let's summarize the main points:

1. **Arithmetic Sequences:**
  - Defined by a constant difference  $d$ .
  - The  $n$ th term:  $u_n = a + (n - 1)d$ .
2. **Arithmetic Series:**
  - The sum of the first  $n$  terms is:

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l).$$

3. **Geometric Sequences:**
  - Each term is multiplied by a constant ratio  $r$ .
  - The  $n$ th term:  $u_n = a r^{n-1}$ .

4. **Geometric Series:**

- The sum of the first  $n$  terms is:

$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$$

5. **Sum to Infinity:**

- For  $|r| < 1$ , the sum to infinity is:

$$S_\infty = \frac{a}{1 - r}.$$

6. **Sigma Notation:**

- A compact form to express sums; for example,  $\sum_{i=1}^n a_i$ .

7. **Recurrence Relations:**

- Define sequences recursively; these can often be translated into explicit formulas.

8. **Modelling with Series:**

- Real-world problems, such as savings plans or compound interest, can be modelled using arithmetic or geometric series.

## SUMMARY OF KEY POINTS

- **Arithmetic sequences** have constant differences; their  $n$ th term is  $a + (n - 1)d$ .
- **Arithmetic series** sums can be found by the formula  $S_n = \frac{n}{2}(2a + (n - 1)d)$  or  $S_n = \frac{n}{2}(a + l)$ .
- **Geometric sequences** are defined by a common ratio; the  $n$ th term is  $u_n = ar^{n-1}$ .
- **Geometric series** sum to  $S_n = \frac{a(1-r^n)}{1-r}$ ; the infinite sum exists for  $|r| < 1$  and equals  $\frac{a}{1-r}$ .
- **Sigma notation** succinctly expresses sums.
- **Recurrence relations** can define sequences recursively, and knowing how to convert them to explicit formulas is useful.
- **Applications:** Both arithmetic and geometric series provide essential models in finance, science, and engineering.

## CHAPTER 6: DIFFERENTIATION

Differentiation is not only about finding a function's derivative; it is also a powerful tool for analyzing how functions behave. In this chapter, we focus on:

1. Determining when functions are increasing or decreasing.
2. Identifying and classifying stationary points.
3. Sketching the graph of the gradient (first derivative) from a given function.
4. Modeling using differentiation in contexts such as rates of change and optimization.

## 6.1 Increasing and Decreasing Functions

### A. Using the First Derivative to Determine Monotonicity

- **Definition:** A function  $f(x)$  is said to be **increasing** on an interval if, for any two points  $x_1$  and  $x_2$  in that interval with  $x_1 < x_2$ , we have

$$f(x_1) < f(x_2).$$

Conversely,  $f(x)$  is **decreasing** if  $f(x_1) > f(x_2)$  whenever  $x_1 < x_2$ .

- **Key Principle Using Derivatives:** The first derivative  $f'(x)$  provides the instantaneous rate of change:
  - If  $f'(x) > 0$  **on an interval**, then  $f(x)$  is increasing on that interval.
  - If  $f'(x) < 0$  **on an interval**, then  $f(x)$  is decreasing on that interval.

### B. Examples

1. **Example — Increasing Function:** Consider  $f(x) = x^3$ .
  - Compute its derivative:

$$f'(x) = 3x^2$$

Notice that  $3x^2 \geq 0$  for all  $x$  (with  $f'(x) = 0$  only at  $x=0$ ). Thus, apart from the stationary point at 0, the function is increasing everywhere since when  $x > 0$  and  $x < 0$  (keeping in mind the shape of a cubic), the function's trend is upward (strictly increasing for  $x > 0$ , and increasing in the sense that the output becomes less negative as  $x$  increases from negative values to 0).

2. **Example — Decreasing Function:** Consider  $f(x) = -2x + 5$ .
  - Its derivative is:

$$f'(x) = -2.$$

Since  $-2 < 0$  for all  $x$ , the function is decreasing on the entire real line.

3. **Application:** For a less trivial function, suppose  $f(x) = x^2 - 4x + 3$ .
  - Its derivative:

$$f'(x) = 2x - 4.$$

- Set  $f'(x) = 0$  to find the turning point:

$$2x - 4 = 0 \quad \Rightarrow \quad x = 2.$$

- For  $x < 2$ , choose  $x=0$ :  $f'(0) = -4 < 0$ , so  $f(x)$  is decreasing.
- For  $x > 2$ , choose  $x=4$ :  $f'(4) = 4 > 0$  so  $f(x)$  is increasing.
- Hence, the function decreases for  $x < 2$  and increases for  $x > 2$ .

## 6.2 Stationary Points

### A. Understanding Stationary Points

- **Definition:** A **stationary point** is a point on the graph of  $f(x)$  where the derivative is zero:

$$f'(a) = 0.$$

These points are candidates for local maxima, local minima, or (in some cases) points of inflection.

### B. Using the First and Second Derivatives to Classify Stationary Points

- **Classification via the Second Derivative Test:**
  - If at  $x = a$ ,  $f'(a) = 0$  and  $f''(a) > 0$ , then  $f(x)$  is concave upward near  $a$  and there is a **local minimum**.
  - If at  $x = a$ ,  $f'(a) = 0$  and  $f''(a) < 0$ , then  $f(x)$  is concave downward near  $a$  and there is a **local maximum**.
  - If  $f'(a) = 0$  and  $f''(a) = 0$ , the test is inconclusive; further investigation (such as higher-order derivatives or sign analysis of  $f'(x)$ ) is needed to classify the point.

### C. Examples

1. **Example — Local Minimum:** Consider  $f(x) = x^2$ .
  - Its derivative  $f'(x) = 2x$  is zero at  $x = 0$ .
  - The second derivative  $f''(x) = 2$  (which is positive) indicates a local minimum at  $x=0$ .
2. **Example — Local Maximum:** Consider  $f(x) = -x^2$ .
  - Here,  $f'(x) = -2x$  and  $f'(0) = 0$ .
  - The second derivative  $f''(x) = -2$  (which is negative) shows that there is a local maximum at  $x = 0$ .
3. **Example — Inconclusive Second Derivative:** Consider  $f(x) = x^3$ .
  - Its derivative  $f'(x) = 3x^2$  is zero at  $x = 0$ .
  - However,  $f''(x) = 6x$  and  $f''(0) = 0$ .
  - In this case, further investigation (by analyzing the behavior of  $f(x)$  or using the first derivative test) reveals that  $x = 0$  is an inflection point rather than a local maximum or minimum.

## 6.3 Sketching Gradient Functions

### A. Relationship Between $f(x)$ and $f'(x)$

- **Concept:** The graph of the derivative  $f'(x)$  reveals the rate at which the original function  $f(x)$  is changing. Key observations include:
  - **Stationary Points:** When  $f(x)$  has a stationary point (where  $f'(x) = 0$ ), the graph of  $f'(x)$  crosses or touches the x-axis.
  - **Increasing/Decreasing Behavior:** When  $f(x)$  is increasing,  $f'(x) > 0$ ; when  $f(x)$  is decreasing,  $f'(x) < 0$ .
  - **Steepness:** The magnitude of  $f'(x)$  indicates the steepness of  $f(x)$ .

### B. Sketching Process

1. **Identify Critical Points:** Determine points where  $f'(x) = 0$  (stationary points) and note their x-coordinates.
2. **Analyze the Sign of  $f'(x)$ :** Use the increasing/decreasing behavior of  $f(x)$  (from earlier analysis) to determine where  $f'(x)$  is positive or negative.
3. **Relate Curvature:** The behavior of  $f''(x)$  informs whether  $f'(x)$  is increasing (if  $f''(x) > 0$ ) or decreasing (if  $f''(x) < 0$ ).
4. **Draw the Graph:** Start with known points (where  $f'(x) = 0$ ) and sketch the curve above or below the x-axis according to the sign.

### C. Example

Consider  $f(x) = x^2$ :

- We know  $f'(x) = 2x$ .
- The derivative is zero at  $x=0$ , negative when  $x<0$  (since  $2x < 0$ ), and positive when  $x>0$ .
- Thus, the graph of  $f'(x) = 2x$  is a straight line passing through the origin with a positive slope.
- **Interpretation:**
  - When  $f(x)$  is decreasing (for  $x<0$ ),  $f'(x)$  is negative.
  - When  $f(x)$  is increasing (for  $x>0$ ),  $f'(x)$  is positive.

## CHAPTER REVIEW 6

Let's briefly review the main points from this chapter:

1. **Increasing and Decreasing Functions:**
  - The first derivative  $f'(x)$  determines whether a function is increasing ( $f'(x) > 0$ ) or decreasing ( $f'(x) < 0$ ) on an interval.
2. **Stationary Points:**
  - A stationary point occurs when  $f'(x) = 0$ .
  - The second derivative test classifies these points:

- $f''(a) > 0$  implies a local minimum.
- $f''(a) < 0$  implies a local maximum.
- $f''(a) = 0$  requires further investigation.

### 3. Sketching Gradient Functions:

- The graph of  $f'(x)$  reflects the behavior of  $f(x)$ : where  $f(x)$  is increasing,  $f'(x)$  is positive; where  $f(x)$  is decreasing,  $f'(x)$  is negative.
- Stationary points in  $f(x)$  appear as roots of  $f'(x)$ .

## SUMMARY OF KEY POINTS

- **Monotonicity:** Use the sign of  $f'(x)$  to determine if  $f(x)$  is increasing or decreasing.
- **Stationary Points:** Identify points where  $f'(x)=0$  and classify them using  $f''(x)$ .
- **Graphical Insight:** The graph of  $f'(x)$  provides a picture of the function's behavior (e.g., slopes, turning points).

## Chapter 7: Integration

Integration is a central concept in calculus. It not only provides a way to recover functions from their derivatives but also offers a powerful tool for computing areas under curves and between curves. In this chapter, we will examine definite integrals, how they represent area, and several techniques for approximating or exactly calculating areas in various scenarios.

### 7.1 Definite Integrals

*Understanding the Definite Integral as the Area Under a Curve*

- **Concept Overview:** The definite integral of a function  $f(x)$  from  $x = a$  to  $x = b$ ,

$$\int_a^b f(x) \, dx,$$

represents the net area between the graph of  $f(x)$  and the  $x$ -axis over the interval  $[a, b]$ .

- **Positive regions:** When  $f(x) \geq 0$ , the integral adds area above the  $x$ -axis.
- **Negative regions:** When  $f(x) < 0$ , the area below the  $x$ -axis contributes a negative value.
- **Visualization:** Imagine sketching the function  $f(x)$  on an  $x$ - $y$  plane. The area (or net area) is the sum of little rectangles or “strips” under the curve. When these strips fall below the  $x$ -axis, they subtract from the total.
- **Example 1:** Compute the area under  $f(x) = x + 1$  between  $x = 0$  and  $x = 3$ .
  1. **Graph Insight:** The line is entirely above the  $x$ -axis.
  2. **Set Up the Integral:**  $\int_0^3 (x + 1) \, dx$

3. **Find the Indefinite Integral:**

$$\int (x + 1) \, dx = \frac{x^2}{2} + x + C.$$

4. **Apply the Fundamental Theorem (see below for details):**

$$\left[ \frac{x^2}{2} + x \right]_0^3 = \left( \frac{9}{2} + 3 \right) - (0 + 0) = \frac{9}{2} + 3 = \frac{15}{2}.$$

5. **Interpretation:** The area under the curve is  $\frac{15}{2}$  square units.

*Evaluating Definite Integrals Using the Fundamental Theorem of Calculus*

- **The Fundamental Theorem Explained:** If  $F(x)$  is any antiderivative (indefinite integral) of  $f(x)$ , that is,  $F'(x) = f(x)$ , then the definite integral is given by:

$$\int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a).$$

- **Why It Works:** The theorem bridges differentiation and integration. The antiderivative  $F(x)$  “accumulates” all the changes of  $f(x)$ , and by evaluating the difference  $F(b)-F(a)$ , we capture the net change (or net area) from  $a$  to  $b$ .
- **Example 2:** Evaluate

$$\int_1^3 3x^2 \, dx.$$

1. **Find an Antiderivative:**

$$F(x) = x^3 \quad \left( \text{since } \frac{d}{dx}[x^3] = 3x^2 \right).$$

2. **Apply the Fundamental Theorem:**

$$[F(x)]_1^3 = F(3) - F(1) = 3^3 - 1^3 = 27 - 1 = 26. \quad [F(x)]_1^3 = F(3) - F(1) = 3^3 - 1^3 = 27 - 1 = 26.$$

## 8.2 Areas Under Curves

*Finding the Area Between a Curve and the x-Axis Using Definite Integration*

- **The Concept of “Area” Going Beyond Algebraic Sum:** When computing

$$\int_a^b f(x) \, dx,$$

the result represents the net area. However, when we speak of “area,” we typically want a positive quantity.

- **Steps When  $f(x)$  is Non-negative:**
  - If  $f(x) \geq 0$  on  $[a, b]$ , then

$$\text{Area} = \int_a^b f(x) \, dx.$$

- **Example 3:** Find the area under  $f(x) = \sqrt{x}$  from  $x = 0$  to  $x = 4$ .
  1. **Write the Integral:**

$$A = \int_0^4 \sqrt{x} \, dx.$$

2. **Find the Antiderivative:**  $\sqrt{x} = x^{1/2}$  so,

$$\int x^{1/2} \, dx = \frac{2}{3} x^{3/2} + C.$$

3. **Evaluate:**

$$\left[ \frac{2}{3} x^{3/2} \right]_0^4 = \frac{2}{3} (4^{3/2}) - 0 = \frac{2}{3} (8) = \frac{16}{3}.$$

4. **Conclusion:** The area under  $\sqrt{x}$  from 0 to 4 is  $\frac{16}{3}$  square units.

*Dealing with Areas Below the x-Axis*

- **The Issue:** When a function  $f(x)$  is negative over an interval, its definite integral gives a negative number, which represents “net area.” However, if we want the actual geometric area, we must address the sign.
- **How to Proceed:**
  - Compute the integral exactly as defined.
  - Recognize that a negative result indicates the area lies below the  $x$ -axis.
  - **For geometric area, take the absolute value:**

$$\text{Actual area} = \left| \int_a^b f(x) \, dx \right|.$$

- **Example 4:** Suppose  $f(x) = -\sin x$  is integrated from  $x = 0$  to  $x = \pi$ .

1. **Integral Calculation:**

$$\int_0^{\pi} (-\sin x) dx = [\cos x]_0^{\pi} = \cos \pi - \cos 0 = (-1) - 1 = -2.$$

2. **Interpreting the Result:** The negative value indicates the net area is below the xx-axis.  
3. **Actual Area (Absolute Value):**

$$\{\text{Area}\} = |-2| = 2 \quad \{\text{square units}\}.$$

### 7.3 Areas Under the x-Axis

*Calculating Areas Below the x-Axis by Taking the Absolute Value of the Definite Integral*

- **Detailed Explanation:** When a function  $f(x)$  lies entirely below the x-axis, the computed definite integral will be negative. However, when we talk about “area,” we mean the non-negative size of the region.
- **Adjustment Approach:** If  $\int_a^b f(x) dx = -K$  (with  $K > 0$ ), then the actual area is  $K$  (i.e.,  $|-K| = K$ ).
- **Example 5:** Find the area enclosed by  $f(x) = -2x$  on  $[0, 3]$ , which is entirely below the x-axis (for  $x > 0$ ).

1. **Compute the Integral:**

$$\left[ \int_0^3 (-2x) dx = -2 \int_0^3 x dx = -2 \left[ \frac{x^2}{2} \right]_0^3 = -2 \left( \frac{9}{2} \right) = -9. \right]$$

2. **Correct to Positive Area:**

$$\text{Area} = |-9| = 9.$$

3. **Interpretation:** Even though the integral gave -9, the geometric area is 9 square units.

### 7.4 Areas Between Curves

*Finding the Area Between Two Curves*

- **Basic Idea:** When two functions  $f(x)$  and  $g(x)$  share the same domain and are arranged vertically (one above the other), the area of the region between them is given by the integral of their difference:

$$\text{Area} = \int_a^b |f(x) - g(x)| dx.$$

More commonly, if you know which function is on top (say  $f(x) \geq g(x)$  on  $[a, b]$ ), then:

$$\text{Area} = \int_a^b (f(x) - g(x)) \, dx.$$

- **Determining Limits of Integration:** The limits  $a$  and  $b$  are the  $x$ -coordinates where the curves intersect. To find these, set  $f(x) = g(x)$  and solve for  $x$ .
- **Example 6:** Find the area between  $f(x) = 4 - x^2$  (a downward opening parabola) and  $g(x) = x$  (a straight line).
  1. **Find Points of Intersection:** Set  $4 - x^2 = x$  which rearranges to:

$$x^2 + x - 4 = 0.$$

Solve using the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1 + 16}}{2} = \frac{-1 \pm \sqrt{17}}{2}.$$

Assume that on the interval of interest, say from  $a = \frac{-1 - \sqrt{17}}{2}$  to  $b = \frac{-1 + \sqrt{17}}{2}$ ,  $f(x)$  is above  $g(x)$ . (In practice you may restrict to the part where the curves make sense for the problem.)

2. **Express the Area:** If  $f(x) \geq g(x)$  over  $[a, b]$ , then:

$$\text{Area} = \int_a^b [(4 - x^2) - x] \, dx.$$

3. **Evaluate the Integral:** Write the integrand as:

$$4 - x^2 - x.$$

Find its antiderivative:

$$4x - \frac{x^3}{3} - \frac{x^2}{2}.$$

Then apply the limits (the exact numerical evaluation will depend on the computed intersection points).

## 7.5 Areas Between Two Curves and Lines

*Calculating Areas Bounded by Curves and Straight Lines*

- **Extended Concept:** Sometimes, the region in question isn't bounded by two curves alone; one or more boundaries might be straight lines (for example, vertical lines  $x=a$  or  $x=b$ , or horizontal lines  $y=c$ ).
- **Approach:**

1. **Identify the Boundaries:** Determine which functions (curves or lines) form the boundary.
2. **Determine the Integration Limits:** Often, vertical or horizontal lines give the limits of integration directly.
3. **Set Up the Integral:**
  - For vertical cross-sections, if the top boundary is  $f(x)$  and the bottom boundary is  $g(x)$ , then:

$$\text{Area} = \int_{x=a}^{x=b} (f(x) - g(x)) \, dx.$$

- For horizontal cross-sections (when integrating with respect to  $y$ ), you would express the curves as  $x = h(y)$  and adjust the limits accordingly.
- **Example 7:** Find the area of the region bounded by the parabola  $y = x^2$ , the line  $y = 4$ , and the vertical line  $x = -2$  (assuming symmetry gives another boundary on the right).
  1. **Determine Intersection Points:**
    - The line  $y=4$  meets  $y = x^2$  when  $x^2 = 4$  so  $x = \pm 2$ .
  2. **Set Up the Integral:**
    - The region between the parabola and line  $y = 4$  from  $x = -2$  to  $x = 2$  has area:

$$\text{Area} = \int_{-2}^2 (4 - x^2) \, dx.$$

3. **Evaluate the Integral:**
  - Find the antiderivative:

$$4x - \frac{x^3}{3}.$$

- Use symmetry if applicable (since the function is even with respect to  $x$  if rearranged properly):

$$\text{Area} = 2 \int_0^2 (4 - x^2) \, dx.$$

- Completing the evaluation yields the total area.

## 7.6 The Trapezium Rule

*Using the Trapezium Rule to Approximate the Area Under a Curve*

- **Overview:** The trapezium rule (or trapezoidal rule) is a numerical method used to approximate the definite integral (and hence the area under a curve). Instead of summing infinitely many infinitely thin rectangles (as in a Riemann sum), we approximate the area by summing areas of trapezoids.

- **Formula:** Suppose you want to approximate

$$\int_a^b f(x) \, dx,$$

and you divide the interval into  $n$  equal subintervals of width

$$h = \frac{b - a}{n}.$$

Then the trapezium rule gives:

$$T = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)],$$

where  $x_i = a + i h$ .

- **Understanding Overestimate vs. Underestimate:**
  - **Concavity Matters:** The error in the trapezium rule is related to the second derivative  $f''(x)$ .
    - If  $f(x)$  is **concave up** (i.e.,  $f''(x) > 0$ ), the trapezium rule tends to **underestimate** the true area because the tops of the trapezoids lie below the actual curve.
    - If  $f(x)$  is **concave down** (i.e.,  $f''(x) < 0$ ), it tends to **overestimate** the area.
- **Example 8:** Approximate the area under  $f(x) = \cos x$  on  $\left[0, \frac{\pi}{2}\right]$  using 4 subintervals.

1. **Calculate  $h$ :**

$$h = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}.$$

2. **Determine the points:**

3. **Apply the Trapezium Rule:**

$$T = \frac{h}{2} \left[ f(0) + 2f\left(\frac{\pi}{8}\right) + 2f\left(\frac{\pi}{4}\right) + 2f\left(\frac{3\pi}{8}\right) + f\left(\frac{\pi}{2}\right) \right].$$

Substitute the known values (numerical approximations may be used for  $\cos(\pi/8)$  and  $\cos(3\pi/8)$ ):

$$T = \frac{\pi/8}{2} \left[ 1 + 2 \cos\left(\frac{\pi}{8}\right) + 2 \left(\frac{\sqrt{2}}{2}\right) + 2 \cos\left(\frac{3\pi}{8}\right) + 0 \right].$$

Simplify:

$$T = \frac{\pi}{16} \left[ 1 + 2 \cos\left(\frac{\pi}{8}\right) + \sqrt{2} + 2 \cos\left(\frac{3\pi}{8}\right) \right].$$

Using approximate values:  $\cos\left(\frac{\pi}{8}\right) \approx 0.9239$  –  $\cos\left(\frac{3\pi}{8}\right) \approx 0.3827$

We get:

$$T \approx \frac{\pi}{16} [1 + 2(0.9239) + 1.4142 + 2(0.3827)].$$

Inside the bracket:  $1 + 1.8478 + 1.4142 + 0.7654 \approx 5.0274$ .

So:

$$T \approx \frac{\pi}{16} \times 5.0274 \approx 0.19635 \times 5.0274 \approx 0.987.$$

#### 4. Analysis of Over/Underestimate:

- Since  $f(x) = \cos x$  is concave down on  $[0, \pi/2]$  (its second derivative  $-\cos x$  is negative in this interval), the trapezium rule is expected to give a slightly **overestimated** area compared to the true value.

## Chapter Review 7

Let's briefly review what we have seen:

### 1. Definite Integrals:

- They measure net area under a curve.
- The Fundamental Theorem of Calculus bridges the derivative and the integral.

### 2. Areas Under Curves:

- When a function is entirely above the x-axis, the definite integral equals the area.
- When  $f(x)$  is below, the integral is negative, and the absolute value gives the "true" area.

### 3. Areas Under the x-Axis:

- A special case of the above, emphasizing the need to take the absolute value to report a positive area.

### 4. Areas Between Curves:

- The area between two curves is found by integrating the difference  $f(x) - g(x)$ , ensuring correct ordering (top minus bottom).
- Intersection points determine the integration limits.

### 5. Areas Between Curves and Lines:

- The same principles apply but with straight lines as additional boundaries.
- Correctly identifying integration limits is key to setting up the right integral.

### 6. The Trapezium Rule:

- A numerical approximation method.
- The formula averages the endpoints and doubles the intermediate values.

- Its accuracy depends on the concavity of the function.